

Asymptotic Fisher information for Poisson counts driven by a stationary Gaussian process

Mankei Tsang^{1,2,*}

¹*Department of Electrical and Computer Engineering,
National University of Singapore, 4 Engineering Drive 3, Singapore 117583*

²*Department of Physics, National University of Singapore, 2 Science Drive 3, Singapore 117551*

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This memo rigorously proves Eq. (3.17) in Ng *et al.*, Phys. Rev. A **93**, 042121 (2016) for the asymptotic Fisher information of spectral photon counting. Let X be a real zero-mean stationary Gaussian process with spectral density S_θ , and let the observations be conditionally independent Poisson random variables whose means are the squared moduli of the positive-frequency discrete Fourier coefficients of a length- T sample of X . Under Wiener-algebra regularity and strict positivity of the spectrum, we prove

$$\lim_{T \rightarrow \infty} \frac{I_T(\theta)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{[\partial_\theta S_\theta(\omega)]^2}{S_\theta(\omega)[1 + S_\theta(\omega)]} d\omega.$$

ChatGPT 5.6 Sol produced the proof, following a suggestion by Tsang. Claude Opus 5.5 checked the proof and proposed minor improvements.

I. INTRODUCTION

The goal of this work is to rigorously prove Eq. (3.17) in Ref. [1] for the asymptotic Fisher information of spectral photon counting, after a zero-mean stationary Gaussian process with a parameter-dependent spectral density modulates a coherent optical beam.

Ref. [1] derives the formula by showing that the photon counts tend to a set of independent Bose–Einstein (geometric) random variables in the long-time limit and then computing the Fisher information of the Bose–Einstein variables. However, ChatGPT complained that this approach is not rigorous, as one should instead take the long-time limit of the Fisher information to determine the asymptotic Cramér–Rao bound; taking the long-time limit prematurely need not give the correct result.

Following Refs. [1, 2], Tsang suggested to ChatGPT the general strategy of sandwiching the Fisher information between a lower bound due to Stein, Mezghani, and Nossek [3] and an upper bound due to Zamir [4], as he suspected that the two bounds, once optimized, should give the same asymptotic formula. ChatGPT took the suggestion and eventually produced the proof below, after a few rounds of encouragement by Tsang. The proof was then bounced between Claude and ChatGPT for checking and improvements. Both eventually claimed the proof to be correct, although Tsang has not checked it independently.

II. MODEL AND ASSUMPTIONS

Identify $\mathbb{T}$ with $[-\pi, \pi)$ and let $A(\mathbb{T})$ denote the Wiener algebra

$$A(\mathbb{T}) = \left\{ a(\omega) = \sum_{k \in \mathbb{Z}} a_k e^{-ik\omega} : \sum_{k \in \mathbb{Z}} |a_k| < \infty \right\}, \quad \|a\|_A = \sum_k |a_k|.$$

Fix a parameter value θ_0 . Assume that on an open interval Θ containing θ_0 :

* mankei@nus.edu.sg; <https://blog.nus.edu.sg/mankei/>

(A1) $\theta \mapsto S_\theta$ is C^1 as a map $\Theta \rightarrow A(\mathbb{T})$;

(A2) S_θ is real and even, and there are constants $0 < m < M < \infty$ such that

$$m \leq S_\theta(\omega) \leq M, \quad \theta \in \Theta, \omega \in \mathbb{T}.$$

No separate likelihood-regularity assumption is imposed: the finite-dimensional Gaussian and Gaussian–Poisson models used below are shown in Appendix A to possess the score identities and finite Fisher information required by the proof.

Let $X = (X_t)_{t \in \mathbb{Z}}$ be a real, zero-mean stationary Gaussian process with spectral density S_θ . Thus

$$C_\theta(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_\theta(\omega) e^{ik\omega} d\omega, \quad S_\theta(\omega) = \sum_{k \in \mathbb{Z}} C_\theta(k) e^{-ik\omega}.$$

For $T \geq 3$ let $\omega_j = 2\pi j/T$ and let F_T be the unitary DFT matrix

$$(F_T)_{jt} = T^{-1/2} e^{-i\omega_j t}, \quad j = 0, \dots, T-1, \quad t = 1, \dots, T.$$

For $X^{(T)} = (X_1, \dots, X_T)^\top$ put $\check{X}_T = F_T X^{(T)}$, that is,

$$\check{X}_{T,j} = \frac{1}{\sqrt{T}} \sum_{t=1}^T X_t e^{-i\omega_j t}, \quad j = 0, \dots, T-1,$$

and let $J_T = \{j : 0 < \omega_j < \pi\}$. Conditional on $X^{(T)}$, the observations are independent and

$$N_{T,j} \mid X^{(T)} \sim \text{Pois}(|\check{X}_{T,j}|^2), \quad j \in J_T. \quad (2.1)$$

Write $I_T(\theta)$ for the Fisher information in $N_T = (N_{T,j})_{j \in J_T}$.

Theorem 1. *Under (A1)–(A2),*

$$\boxed{\lim_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{\dot{S}(\omega)^2}{S(\omega)[1 + S(\omega)]} d\omega,} \quad (2.2)$$

where $S = S_{\theta_0}$ and $\dot{S} = \partial_\theta S_\theta|_{\theta_0}$.

Theorem 1 leads to Eq. (3.17) in Ref. [1] after some trivial rescaling.

We write $\mathcal{I}$ for the right-hand side of (2.2). Throughout, a dot denotes ∂_θ , and quantities written without a θ subscript are evaluated at θ_0 . Theorem 1 follows from Proposition 1 (Section IV) and Proposition 2 (Section V).

III. TOEPLITZ–CIRCULANT LEMMAS

For $a \in A(\mathbb{T})$ define the Toeplitz matrix $\text{Toe}_T(a) = (a_{r-s})_{r,s=1}^T$, so that $\text{Cov}(X^{(T)}) = \text{Toe}_T(S_\theta)$. The map $a \mapsto \text{Toe}_T(a)$ is linear, $\text{Toe}_T(1) = I$, and $\text{Toe}_T(a)$ is Hermitian when a is real. For $v \in \mathbb{C}^T$,

$$v^\dagger \text{Toe}_T(a) v = \frac{1}{2\pi} \int_{-\pi}^{\pi} a(\omega) \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega = \|v\|^2, \quad (3.1)$$

so $(\min a)I \leq \text{Toe}_T(a) \leq (\max a)I$ for real a . In particular, $mI \leq \text{Toe}_T(S_\theta) \leq MI$ by (A2).

For a bounded function a on $\mathbb{T}$ define the sampled circulant

$$\text{Cir}_T(a) = F_T^\dagger \text{diag}(a(\omega_0), \dots, a(\omega_{T-1})) F_T.$$

Sampled circulants are simultaneously diagonalized by F_T , so $\text{Cir}_T(a) \text{Cir}_T(b) = \text{Cir}_T(ab)$, $\text{Cir}_T(a)^\dagger = \text{Cir}_T(a^*)$, $\|\text{Cir}_T(a)\|_{\text{op}} = \max_j |a(\omega_j)|$, and $f(\text{Cir}_T(a)) = \text{Cir}_T(f \circ a)$ for every function f defined on the values $a(\omega_j)$; in particular $\text{Cir}_T(a)^{-1} = \text{Cir}_T(1/a)$ if a does not vanish [5, Theorem 7 and Lemma 10].

For $T \times T$ matrices set $\|M\|_T = T^{-1/2} \|M\|_{\text{HS}}$, where $\|\cdot\|_{\text{HS}}$ denotes the (unnormalized) Hilbert–Schmidt norm, also called the Frobenius norm. Both $\|\cdot\|_{\text{HS}}$ and $\|\cdot\|_T$ are unitarily invariant, and for all matrices M, M' and every orthogonal projection P ,

$$\begin{aligned} \|M\|_T &\leq \|M\|_{\text{op}}, & \frac{|\text{tr}(PM)|}{T} &\leq \|M\|_T, \\ \|MM'\|_T &\leq \min\{\|M\|_T \|M'\|_{\text{op}}, \|M\|_{\text{op}} \|M'\|_T\}. \end{aligned} \quad (3.2)$$

(The trace bound follows from $|\text{tr}(PM)| \leq \|P\|_{\text{HS}} \|M\|_{\text{HS}}$ and $\|P\|_{\text{HS}} \leq T^{1/2}$.) We write $M_T \sim \widetilde{M}_T$ if $\|M_T - \widetilde{M}_T\|_T \rightarrow 0$ and both sequences have uniformly bounded operator norm [5]. Finally, for continuous g on $\mathbb{T}$ the Riemann-sum theorem gives

$$\frac{1}{T} \sum_{j=0}^{T-1} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\omega) d\omega, \quad \frac{1}{T} \sum_{j \in J_T} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_0^\pi g(\omega) d\omega. \quad (3.3)$$

Lemma 1 (Toeplitz–circulant equivalence [5, Lemma 11]). *If $a \in A(\mathbb{T})$, then $\text{Toe}_T(a) \sim \text{Cir}_T(a)$. Moreover, both $\|\text{Toe}_T(a)\|_{\text{op}}$ and $\|\text{Cir}_T(a)\|_{\text{op}}$ are at most $\|a\|_A$.*

Lemma 2 (Stability [5, Theorem 1]). *Suppose $M_T \sim \widetilde{M}_T$ and $M'_T \sim \widetilde{M}'_T$. Then $M_T + M'_T \sim \widetilde{M}_T + \widetilde{M}'_T$, $M_T^\dagger \sim \widetilde{M}_T^\dagger$, and $M_T M'_T \sim \widetilde{M}_T \widetilde{M}'_T$. If, in addition, M_T and $\widetilde{M}_T$ are invertible and their inverses have uniformly bounded operator norm, then $M_T^{-1} \sim \widetilde{M}_T^{-1}$.*

Lemma 3 (Functional calculus for Toeplitz–circulant sequences). *Let $\theta \mapsto a_\theta \in A(\mathbb{T})$ be C^1 in a neighborhood of θ_0 , with a_θ real-valued, and suppose that for some fixed $\alpha < \beta$,*

$$\alpha \leq a_\theta(\omega) \leq \beta$$

for all ω and all θ in that neighborhood. Write

$$a = a_{\theta_0}, \quad \dot{a} = \partial_\theta a_\theta|_{\theta=\theta_0}.$$

Let f be analytic on an open neighborhood of $[\alpha, \beta]$. Then

$$f(\text{Toe}_T(a)) \sim \text{Cir}_T(f(a)), \quad (3.4)$$

and

$$\left. \frac{d}{d\theta} f(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} \sim \text{Cir}_T(f'(a)\dot{a}). \quad (3.5)$$

Proof. By Lemma 1,

$$\text{Toe}_T(a) \sim \text{Cir}_T(a), \quad \text{Toe}_T(\dot{a}) \sim \text{Cir}_T(\dot{a}).$$

Since Toe_T is linear with $\|\text{Toe}_T(b)\|_{\text{op}} \leq \|b\|_A$, the map $\theta \mapsto \text{Toe}_T(a_\theta)$ is C^1 in operator norm, with derivative $\text{Toe}_T(\dot{a})$ at θ_0 ; by Eq. (3.1), $\text{Toe}_T(a_\theta)$ is Hermitian with spectrum in $[\alpha, \beta]$ for all θ in the neighborhood. Hence, for every function g analytic on a neighborhood of $[\alpha, \beta]$, the chain rule gives

$$\left. \frac{d}{d\theta} g(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} = Dg(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})],$$

where $Dg(A)[E]$ denotes the Fréchet derivative of $A \mapsto g(A)$ at A in the direction E .

We first prove the result for a polynomial

$$p(x) = \sum_{k=0}^d c_k x^k.$$

Repeated application of Lemma 2 gives

$$p(\text{Toe}_T(a)) \sim p(\text{Cir}_T(a)) = \text{Cir}_T(p(a)),$$

where the last equality follows from simultaneous diagonalization of circulant matrices by the DFT.

For the derivative, the chain rule above and the usual product rule give

$$\left. \frac{d}{d\theta} p(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} = Dp(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})] = \sum_{k=1}^d c_k \sum_{\ell=0}^{k-1} \text{Toe}_T(a)^\ell \text{Toe}_T(\dot{a}) \text{Toe}_T(a)^{k-1-\ell}.$$

Again by Lemma 2, this is asymptotically equivalent to

$$\sum_{k=1}^d c_k \sum_{\ell=0}^{k-1} \text{Cir}_T(a)^\ell \text{Cir}_T(\dot{a}) \text{Cir}_T(a)^{k-1-\ell}.$$

The circulant matrices commute, so the latter equals

$$p'(\text{Cir}_T(a)) \text{Cir}_T(\dot{a}) = \text{Cir}_T(p'(a)\dot{a}).$$

Thus both assertions hold for polynomials.

Now let f be analytic on a neighborhood of $[\alpha, \beta]$. There exists a sequence of polynomials p_n such that

$$\sup_{x \in [\alpha, \beta]} (|p_n(x) - f(x)| + |p'_n(x) - f'(x)|) \longrightarrow 0. \quad (3.6)$$

For instance, approximate f' uniformly on $[\alpha, \beta]$ by polynomials q_n , using the Weierstrass theorem for its real and imaginary parts, and set $p_n(x) = f(\alpha) + \int_\alpha^x q_n(t) dt$; then $\|p_n - f\|_{\infty, [\alpha, \beta]} \leq (\beta - \alpha) \|q_n - f'\|_{\infty, [\alpha, \beta]}$.

For the first assertion, since $\text{Toe}_T(a)$ and $\text{Cir}_T(a)$ are Hermitian with spectra contained in $[\alpha, \beta]$,

$$\|f(\text{Toe}_T(a)) - p_n(\text{Toe}_T(a))\|_{\text{op}} \leq \|f - p_n\|_{\infty, [\alpha, \beta]},$$

and likewise for $\text{Cir}_T(a)$. For the same reason, $\|f(\text{Toe}_T(a))\|_{\text{op}}$ and $\|\text{Cir}_T(f(a))\|_{\text{op}} = \|f(\text{Cir}_T(a))\|_{\text{op}}$ are at most $\|f\|_{\infty, [\alpha, \beta]}$ for every T , which gives the operator-norm bounds required by the definition of $\sim$. Since $\|\cdot\|_T \leq \|\cdot\|_{\text{op}}$, the polynomial result followed by $n \rightarrow \infty$ proves Eq. (3.4).

For the differentiated assertion, let $g_n = f - p_n$. If $A = U \text{diag}(\lambda_1, \dots, \lambda_T) U^\dagger$ is Hermitian with spectrum in $[\alpha, \beta]$, the divided-difference formula for the Fréchet derivative gives

$$Dg_n(A)[E] = U(g_n^{[1]}(\lambda_i, \lambda_j) \circ (U^\dagger E U)) U^\dagger;$$

see, for example, Higham [6, Chapter 3]. Hence, by unitary invariance of $\|\cdot\|_T$ and the mean-value theorem,

$$\|Dg_n(A)[E]\|_T \leq \|g'_n\|_{\infty, [\alpha, \beta]} \|E\|_T.$$

Taking $A = \text{Toe}_T(a)$ and $E = \text{Toe}_T(\dot{a})$, and using $\|E\|_T \leq \|E\|_{\text{op}} \leq \|\dot{a}\|_A$, we obtain

$$\sup_T \|Df(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})] - Dp_n(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})]\|_T \longrightarrow 0 \quad (n \rightarrow \infty)$$

by Eq. (3.6).

The operator norm of $Df(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})]$ is also bounded uniformly in T . Fix a contour γ that surrounds $[\alpha, \beta]$ and lies, together with the region it encloses, in the domain of analyticity of f , and let $\delta = \text{dist}(\gamma, [\alpha, \beta]) > 0$. For Hermitian A with spectrum in $[\alpha, \beta]$, the resolvent satisfies $\|(zI - A)^{-1}\|_{\text{op}} \leq \delta^{-1}$ for $z \in \gamma$, so the Cauchy integral representation [6, Chapter 3]

$$Df(A)[E] = \frac{1}{2\pi i} \oint_{\gamma} f(z)(zI - A)^{-1} E (zI - A)^{-1} dz$$

gives

$$\|Df(A)[E]\|_{\text{op}} \leq \frac{\text{length}(\gamma)}{2\pi\delta^2} \sup_{z \in \gamma} |f(z)| \|E\|_{\text{op}}$$

in every dimension. With $A = \text{Toe}_T(a)$ and $E = \text{Toe}_T(\dot{a})$, and since $\sup_T \|\text{Toe}_T(\dot{a})\|_{\text{op}} \leq \|\dot{a}\|_A$, this shows that

$$\sup_T \|Df(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})]\|_{\text{op}} < \infty.$$

For the circulants, simultaneous DFT diagonalization gives directly

$$Df(\text{Cir}_T(a))[\text{Cir}_T(\dot{a})] = \text{Cir}_T(f'(a)\dot{a}),$$

and, by Eq. (3.6),

$$\sup_T \|\text{Cir}_T((p'_n(a) - f'(a))\dot{a})\|_{\text{op}} \longrightarrow 0 \quad (n \rightarrow \infty).$$

Also,

$$\|\text{Cir}_T(f'(a)\dot{a})\|_{\text{op}} \leq \|f'\|_{\infty, [\alpha, \beta]} \|\dot{a}\|_{\infty},$$

so the circulant derivative sequence is uniformly bounded in operator norm. By the chain rule above, the left-hand side of Eq. (3.5) equals $Df(\text{Toe}_T(a))[\text{Toe}_T(\dot{a})]$. Combining these estimates with the polynomial result and then letting $n \rightarrow \infty$ proves Eq. (3.5), including the operator-norm boundedness required by the definition of $\sim$. $\square$

IV. LOWER BOUND

Set

$$\Gamma_T = \text{Toe}_T(S), \quad A_T = F_T \Gamma_T F_T^\dagger, \quad B_T = F_T \Gamma_T F_T^\top.$$

Since $\check{X}_T = F_T X^{(T)}$,

$$(A_T)_{jk} = \mathbb{E}(\check{X}_{T,j} \check{X}_{T,k}^*), \quad (B_T)_{jk} = \mathbb{E}(\check{X}_{T,j} \check{X}_{T,k}).$$

Write $s_{T,j} = (A_T)_{jj} = \mathbb{E}|\check{X}_{T,j}|^2$, and note that $0 < s_{T,j} \leq M$ because $mI \leq \Gamma_T \leq MI$.

The real and imaginary parts of the $\check{X}_{T,j}$ are jointly Gaussian with mean zero, so Isserlis' theorem gives

$$\text{Cov}(|\check{X}_{T,j}|^2, |\check{X}_{T,k}|^2) = |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2.$$

Using conditional independence and the variance formula for a Poisson variable,

$$\text{Cov}(N_{T,j}, N_{T,k}) = \delta_{jk} s_{T,j} + |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2, \quad j, k \in J_T. \quad (4.1)$$

Define

$$w(\omega) = \frac{\dot{S}(\omega)}{S(\omega)[1 + S(\omega)]}, \quad w_j = w(\omega_j).$$

By (A1)–(A2), w is bounded and continuous. At the fixed parameter value θ_0 , define the scalar statistic

$$Q_T = \sum_{j \in J_T} w_j N_{T,j}. \quad (4.2)$$

The weights are held fixed when differentiating in θ .

Proposition 1 (Lower bound). *Under (A1)–(A2), $\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}$.*

Proof. By Lemma A.1 in Appendix A, $\partial_\theta \mathbb{E}_\theta Q_T|_{\theta_0} = \mathbb{E}_{\theta_0}[(Q_T - \mathbb{E}_{\theta_0} Q_T) \text{score}_{N,\theta_0}]$, where $\text{score}_{N,\theta}$ is the score of N_T . The Cauchy–Schwarz inequality therefore yields the scalar information inequality

$$I_T(\theta_0) \geq \frac{[\partial_\theta \mathbb{E}_\theta Q_T|_{\theta_0}]^2}{\text{Var}_{\theta_0}(Q_T)} \quad (4.3)$$

whenever $\text{Var}_{\theta_0}(Q_T) > 0$.

Numerator. We have $\mathbb{E}_\theta Q_T = \sum_{j \in J_T} w_j s_{T,j}(\theta)$ with $s_{T,j}(\theta) = (F_T \text{Toe}_T(S_\theta) F_T^\dagger)_{jj}$, so

$$\partial_\theta \mathbb{E}_\theta Q_T|_{\theta_0} = \sum_{j \in J_T} w_j \dot{s}_{T,j}, \quad \dot{s}_{T,j} = (F_T \text{Toe}_T(\dot{S}) F_T^\dagger)_{jj}.$$

By unitary invariance and Lemma 1 applied to $\dot{S}$,

$$\frac{1}{T} \sum_{j=0}^{T-1} |\dot{s}_{T,j} - \dot{S}(\omega_j)|^2 \leq \|\text{Toe}_T(\dot{S}) - \text{Cir}_T(\dot{S})\|_T^2 \rightarrow 0.$$

Since w is bounded, the Cauchy–Schwarz inequality and (3.3) imply

$$\frac{1}{T} \partial_\theta \mathbb{E}_\theta Q_T|_{\theta_0} \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega) \dot{S}(\omega) d\omega = \mathcal{I}. \quad (4.4)$$

Variance. By Lemma 1 and unitary invariance,

$$\sum_{j,k=0}^{T-1} |(A_T)_{jk} - \delta_{jk} S(\omega_j)|^2 = \|\Gamma_T - \text{Cir}_T(S)\|_{\text{HS}}^2 = o(T). \quad (4.5)$$

Hence

$$\sum_{j \neq k} |(A_T)_{jk}|^2 = o(T), \quad \sum_j |s_{T,j} - S(\omega_j)|^2 = o(T). \quad (4.6)$$

Let Π_T be the permutation matrix of $j \mapsto -j \pmod{T}$. A direct computation gives $F_T F_T^\top = \Pi_T$, hence

$$F_T \text{Cir}_T(S) F_T^\top = \text{diag}(S(\omega_j)) \Pi_T.$$

For $j, k \in J_T$ we have $2 \leq j + k \leq T - 1$, so $k \not\equiv -j \pmod{T}$, and the $J_T \times J_T$ block of this matrix vanishes. As $F_T^\top$ is unitary, Lemma 1 gives

$$\sum_{j, k \in J_T} |(B_T)_{jk}|^2 \leq \|\Gamma_T - \text{Cir}_T(S)\|_{\text{HS}}^2 = o(T). \quad (4.7)$$

By (4.1),

$$\text{Var}(Q_T) = \sum_{j \in J_T} w_j^2 (s_{T,j} + s_{T,j}^2) + \sum_{\substack{j, k \in J_T \\ j \neq k}} w_j w_k |(A_T)_{jk}|^2 + \sum_{j, k \in J_T} w_j w_k |(B_T)_{jk}|^2,$$

and by boundedness of w , (4.6) and (4.7), the last two sums are $o(T)$. Since $|s_{T,j}^2 - S(\omega_j)^2| \leq 2M|s_{T,j} - S(\omega_j)|$, the second relation in (4.6), the Cauchy–Schwarz inequality and (3.3) yield

$$\frac{1}{T} \text{Var}(Q_T) \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega)^2 S(\omega) [1 + S(\omega)] d\omega = \mathcal{I}. \quad (4.8)$$

If $\mathcal{I} > 0$, then $\text{Var}(Q_T) > 0$ for large T , and (4.3), (4.4) and (4.8) imply

$$\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}. \quad (4.9)$$

If $\mathcal{I} = 0$, (4.9) follows from nonnegativity of Fisher information. $\square$

V. UPPER BOUND BY EXACT GAUSSIAN AUGMENTATION

Define, for every $\theta \in \Theta$,

$$R_\theta = 1 + S_\theta, \quad H_\theta = \frac{S_\theta}{1 + S_\theta}. \quad (5.1)$$

Then $S_\theta = R_\theta H_\theta$ and, since $\text{Toe}_T(1) = I$, $\text{Toe}_T(R_\theta) = I + \text{Toe}_T(S_\theta)$. Let

$$Z^{(T)} \sim N(0, \text{Toe}_T(R_\theta))$$

and set

$$L_T(\theta) = \text{Toe}_T(S_\theta)^{1/2} \text{Toe}_T(R_\theta)^{-1/2}, \quad X^{(T)} = L_T(\theta) Z^{(T)}. \quad (5.2)$$

Both factors of $L_T(\theta)$ are functions of the same real symmetric matrix $\text{Toe}_T(S_\theta)$, so

$$L_T(\theta) = h(\text{Toe}_T(S_\theta)), \quad h(x) = \sqrt{\frac{x}{1+x}}. \quad (5.3)$$

In particular $L_T(\theta)$ is real symmetric, and $L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta)$ because $h(x)^2(1+x) = x$. Thus $X^{(T)}$ has exactly the original Gaussian law, and if N_T is generated from $X^{(T)}$ by (2.1), its marginal law is exactly the original law, for every T and every $\theta \in \Theta$. By Lemma A.2 in Appendix A, applied to (5.1),

$$I_T(\theta_0) \leq I_T^Z + \mathbb{E} I_T^{N|Z}(Z^{(T)}), \quad (5.4)$$

where I_T^Z is the Fisher information of $Z^{(T)}$ and $I_T^{N|Z}(z)$ is the conditional Fisher information of N_T given $Z^{(T)} = z$, both at θ_0 .

Proposition 2 (Upper bound). *Under (A1)–(A2), $\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}$.*

Proof. Gaussian term. For a zero-mean real Gaussian vector whose positive definite covariance G_θ is C^1 in θ , the Fisher information is $\frac{1}{2} \text{tr}(G^{-1} \dot{G} G^{-1} \dot{G})$. Therefore

$$I_T^Z = \frac{1}{2} \text{tr}\{\text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R}) \text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R})\}.$$

By Lemma 1, $\text{Toe}_T(R) \sim \text{Cir}_T(R)$ and $\text{Toe}_T(\dot{R}) \sim \text{Cir}_T(\dot{R})$. Since $\text{Toe}_T(R) \geq (1+m)I$ and $\text{Cir}_T(R) \geq (1+m)I$, Lemma 2 gives $\text{Toe}_T(R)^{-1} \sim \text{Cir}_T(1/R)$, and then that the product inside the trace is asymptotically equivalent to $\text{Cir}_T((\dot{R}/R)^2)$. By (3.2) with $P = I$, normalized traces of asymptotically equivalent matrices differ by $o(1)$, so (3.3) and evenness give

$$\lim_{T \rightarrow \infty} \frac{I_T^Z}{T} = \frac{1}{4\pi} \int_{-\pi}^{\pi} \left(\frac{\dot{R}}{R} \right)^2 d\omega = \frac{1}{2\pi} \int_0^\pi (\partial_\theta \log R)^2 d\omega. \quad (5.5)$$

Conditional Poisson term. At θ_0 put

$$\check{X} = F_T L_T Z^{(T)}, \quad V = F_T \dot{L}_T Z^{(T)},$$

where $\dot{L}_T = \partial_\theta L_T(\theta)|_{\theta_0}$. Conditional on $Z^{(T)}$, the Poisson mean at frequency $j \in J_T$ is $\lambda_j = |\check{X}_j|^2$, which is almost surely positive (see Lemma A.2), and, holding $Z^{(T)}$ fixed,

$$\dot{\lambda}_j = 2 \text{Re}(\check{X}_j^* V_j).$$

The conditional Fisher information is therefore

$$I_T^{N|Z}(Z^{(T)}) = \sum_{j \in J_T} \frac{\dot{\lambda}_j^2}{\lambda_j} = 4 \sum_{j \in J_T} \frac{[\text{Re}(\check{X}_j^* V_j)]^2}{|\check{X}_j|^2},$$

and since $[\text{Re}(y^* v)]^2 \leq |y|^2 |v|^2$,

$$I_T^{N|Z}(Z^{(T)}) \leq 4 \sum_{j \in J_T} |V_j|^2. \quad (5.6)$$

We next identify $\dot{L}_T$. Apply Lemma 3 with

$$a_\theta = S_\theta, \quad f(x) = h(x) = \sqrt{\frac{x}{1+x}}.$$

By (A1)–(A2), the hypotheses of the lemma are satisfied with $[\alpha, \beta] = [m, M]$. Since

$$L_T(\theta) = h(\text{Toe}_T(S_\theta)),$$

Eq. (3.5) gives

$$\dot{L}_T \sim \text{Cir}_T(h'(S)\dot{S}) = \text{Cir}_T(\partial_\theta \sqrt{H}). \quad (5.7)$$

Since $\text{Toe}_T(R) \sim \text{Cir}_T(R)$ and $\dot{L}_T$ is real, Lemma 2 gives

$$\dot{L}_T \text{Toe}_T(R) \dot{L}_T^\top \sim \text{Cir}_T(q), \quad q = R(\partial_\theta \sqrt{H})^2 = \frac{1}{4} S(\partial_\theta \log H)^2. \quad (5.8)$$

Let P_T^+ be the diagonal projection in Fourier coordinates onto J_T . Taking expectations in (5.6),

$$\mathbb{E} I_T^{N|Z}(Z^{(T)}) \leq 4 \text{tr}\{P_T^+ F_T \dot{L}_T \text{Toe}_T(R) \dot{L}_T^\top F_T^\dagger\}.$$

By (5.8), unitary invariance and (3.2) with $P = P_T^+$, after division by T the right-hand side differs by $o(1)$ from

$$\frac{4}{T} \text{tr}\{P_T^+ F_T \text{Cir}_T(q) F_T^\dagger\} = \frac{4}{T} \sum_{j \in J_T} q(\omega_j) \longrightarrow \frac{2}{\pi} \int_0^\pi q(\omega) d\omega,$$

where the limit is (3.3). Therefore

$$\limsup_{T \rightarrow \infty} \frac{\mathbb{E} I_T^{N|Z}(Z^{(T)})}{T} \leq \frac{1}{2\pi} \int_0^\pi S(\omega) [\partial_\theta \log H(\omega)]^2 d\omega. \quad (5.9)$$

Combining (5.4), (5.5), and (5.9),

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log R)^2 + S(\partial_\theta \log H)^2\} d\omega. \quad (5.10)$$

For the choice (5.1),

$$\partial_\theta \log R = \frac{\dot{S}}{1+S}, \quad \partial_\theta \log H = \frac{\dot{S}}{S(1+S)},$$

hence pointwise

$$(\partial_\theta \log R)^2 + S(\partial_\theta \log H)^2 = \frac{\dot{S}^2}{(1+S)^2} \left(1 + \frac{1}{S}\right) = \frac{\dot{S}^2}{S(1+S)}, \quad (5.11)$$

and (5.10) becomes

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}. \quad (5.12)$$

□

Proof of Theorem 1. Combine Propositions 1 and 2. □

VI. WHY THE AUGMENTATION IS OPTIMAL

The particular choice (5.1) is not ad hoc.

Definition 1 (Admissible factorization). A family $(R_\theta)_{\theta \in \Theta}$ defines an *admissible factorization* $S_\theta = R_\theta H_\theta$ if it satisfies the hypotheses of Lemma A.2: each R_θ is real and even, $\theta \mapsto R_\theta$ is C^1 from Θ into $A(\mathbb{T})$, and $r_- \leq R_\theta(\omega) \leq r_+$ for constants $0 < r_- \leq r_+ < \infty$. We then put $H_\theta = S_\theta/R_\theta$ and define $Z^{(T)}$ and $L_T(\theta)$ by (5.2).

For an admissible factorization, $L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta)$ still holds, and Lemma A.2 gives (5.4). The matrices $\text{Toe}_T(S_\theta)$ and $\text{Toe}_T(R_\theta)$ need not commute, so (5.3) is not available. Instead, differentiate (5.2) and apply Lemma 3 twice: first with

$$a_\theta = S_\theta, \quad f(x) = x^{1/2},$$

and then with

$$a_\theta = R_\theta, \quad f(x) = x^{-1/2}.$$

The hypotheses hold with the positive intervals $[\alpha, \beta] = [m, M]$ and $[\alpha, \beta] = [r_-, r_+]$, respectively. Together with Lemma 2, the ordinary and differentiated conclusions of Lemma 3 give

$$\dot{L}_T \sim \text{Cir}_T((\partial_\theta \sqrt{S}) R^{-1/2} + \sqrt{S} \partial_\theta R^{-1/2}) = \text{Cir}_T(\partial_\theta \sqrt{H}).$$

The rest of the proof of Proposition 2 applies verbatim, with $1 + m$ replaced by r_- , up to (5.10). Since $\partial_\theta \log R = \partial_\theta \log S - \partial_\theta \log H$, this reads

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log S - \partial_\theta \log H)^2 + S(\partial_\theta \log H)^2\} d\omega.$$

Fix ω and write $a = \partial_\theta \log S_\theta(\omega)|_{\theta_0}$ and $u = \partial_\theta \log H_\theta(\omega)|_{\theta_0}$. The integrand

$$(a - u)^2 + Su^2$$

is strictly convex in u , with unique minimizer $u_* = a/(1 + S)$ and minimum value $a^2 S/(1 + S) = \dot{S}^2/[S(1 + S)]$. Uniqueness refers to the value of $\partial_\theta \log H$ at θ_0 , not to the family: every admissible factorization with $\partial_\theta \log H_\theta|_{\theta_0} = \dot{S}/[S(1 + S)]$ attains the minimum. For instance, let $c \in A(\mathbb{T})$ be positive, even, and independent of θ , and set $R_\theta = (1 + S_\theta)/c$. Since c is continuous and strictly positive on the compact torus, it is bounded above and away from zero; by the Wiener–Lévy theorem, $1/c \in A(\mathbb{T})$. Hence this is an admissible factorization, with $H_\theta = c S_\theta/(1 + S_\theta)$. Taking $c \equiv 1$ gives (5.1). Thus, within this class of augmentations, (5.1) gives the best bound, and that bound meets the moment lower bound exactly.

A one-frequency explanation. Why the two bounds meet can be seen at a single frequency. Heuristically, for $j \in J_T$ the coefficients $\check{Z}_j = (F_T Z^{(T)})_j$ are approximately independent circular complex Gaussians with $\mathbb{E}|\check{Z}_j|^2 \approx R(\omega_j)$, and the Toeplitz–circulant functional calculus gives $L_T \sim \text{Cir}_T(\sqrt{H})$, so that $\check{X}_{T,j} \approx \sqrt{H(\omega_j)} \check{Z}_j$. A single frequency is thus described by

$$E \sim \text{Exp with mean } R, \quad N | E \sim \text{Pois}(HE), \quad HR = S.$$

Since HE is exponential with mean S , N is geometric, $P(N = n) = S^n/(1 + S)^{n+1}$, with Fisher information $\dot{S}^2/[S(1 + S)]$, whatever the factorization. The joint score of (E, N) is

$$\partial_\theta \log R \left(\frac{E}{R} - 1 \right) + (N - HE) \partial_\theta \log H,$$

and its variance $(\partial_\theta \log R)^2 + S(\partial_\theta \log H)^2$ is the integrand of (5.10). For the choice (5.1) the terms involving E cancel, and the joint score reduces to

$$\frac{\dot{S}(N-S)}{S(1+S)} = w(N-S),$$

a function of N alone. This is the score of the geometric law, and it is the centred summand of Q_T . At the level of this limiting one-frequency heuristic, conditioning on N therefore loses no information. This explains the pointwise identity (5.11), while $Q_T - \mathbb{E}Q_T$ is the corresponding sum of per-frequency score-like terms in the finite- T lower-bound construction.

Appendix A: Finite-dimensional likelihood regularity

Fix $T \geq 3$ throughout this appendix. We verify the score identities behind (4.3) and (5.4) directly from (A1)–(A2).

Write $\Gamma_\theta = \text{Toe}_T(S_\theta)$. By (3.1) and (A2), $mI \leq \Gamma_\theta \leq MI$, and since Toe_T is linear with $\|\text{Toe}_T(b)\|_{\text{op}} \leq \|b\|_A$, (A1) implies that $\theta \mapsto \Gamma_\theta$ is C^1 in operator norm. Hence the $N(0, \Gamma_\theta)$ density ϕ_θ is C^1 in θ , with score

$$\text{score}_{X,\theta}(x) = \frac{1}{2} \{x^\top \Gamma_\theta^{-1} \dot{\Gamma}_\theta \Gamma_\theta^{-1} x - \text{tr}(\Gamma_\theta^{-1} \dot{\Gamma}_\theta)\}, \quad \dot{\Gamma}_\theta = \partial_\theta \Gamma_\theta, \quad (\text{A.1})$$

and on every compact subinterval $\Theta_1 \subset \Theta$,

$$\phi_\theta(x) + |\partial_\theta \phi_\theta(x)| \leq C(1 + \|x\|^2) e^{-\|x\|^2/(2M)}, \quad \theta \in \Theta_1, \quad (\text{A.2})$$

for a constant C depending on Θ_1 and T . Consequently, $\theta \mapsto \int g \phi_\theta$ may be differentiated under the integral sign whenever $|g|$ is bounded by a polynomial.

Let $\lambda_j^X(x) = |(F_T x)_j|^2$, the Poisson mean in (2.1) as a function of $X^{(T)} = x$. For $j \in J_T$ the zero set of λ_j^X is $\{x : \sum_t x_t \cos(\omega_j t) = \sum_t x_t \sin(\omega_j t) = 0\}$. Since $0 < \omega_j < \pi$, the vectors $(\cos \omega_j t)_{t=1}^T$ and $(\sin \omega_j t)_{t=1}^T$ are linearly independent, so this zero set is a linear subspace of codimension two and hence Lebesgue-null. The same holds for $z \mapsto \lambda_j^X(Lz)$ with any invertible L .

Lemma A.1 (Original experiment). *Let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For every $\theta \in \Theta$ and $n \in \mathbb{N}_0^{J_T}$, the probability $p_\theta(n) = P_\theta(N_T = n)$ is positive and differentiable in θ , and the score $\text{score}_{N,\theta}(n) = \partial_\theta \log p_\theta(n)$ satisfies*

$$\text{score}_{N,\theta}(N_T) = \mathbb{E}_\theta[\text{score}_{X,\theta}(X^{(T)}) \mid N_T]. \quad (\text{A.3})$$

In particular $\mathbb{E}_\theta \text{score}_{N,\theta} = 0$ and $I_T(\theta) = \mathbb{E}_\theta \text{score}_{N,\theta}^2 \leq \mathbb{E}_\theta \text{score}_{X,\theta}^2 < \infty$. Moreover,

$$\partial_\theta \mathbb{E}_\theta Q_T = \mathbb{E}_\theta[(Q_T - \mathbb{E}_\theta Q_T) \text{score}_{N,\theta}]. \quad (\text{A.4})$$

Proof. The conditional law of N_T given $X^{(T)} = x$ is

$$K_T(n \mid x) = \prod_{j \in J_T} e^{-\lambda_j^X(x)} \frac{\lambda_j^X(x)^{n_j}}{n_j!} \in [0, 1],$$

which does not depend on θ and is positive off a null set. As $\phi_\theta > 0$, $p_\theta(n) = \int K_T(n \mid x) \phi_\theta(x) dx > 0$, and by (A.2)

$$\dot{p}_\theta(n) = \int K_T(n \mid x) \phi_\theta(x) \text{score}_{X,\theta}(x) dx.$$

Dividing by $p_\theta(n)$ gives (A.3). Hence $\mathbb{E}_\theta \text{score}_{N,\theta} = \mathbb{E}_\theta \text{score}_{X,\theta} = 0$, and conditional Jensen gives the bound on $I_T(\theta)$.

For (A.4), the function $g(x) = \mathbb{E}(Q_T \mid X^{(T)} = x) = \sum_{j \in J_T} w_j \lambda_j^X(x)$ is a quadratic form which does not depend on θ . By (A.2) and the tower property,

$$\partial_\theta \mathbb{E}_\theta Q_T = \partial_\theta \int g \phi_\theta = \int g \phi_\theta \text{score}_{X,\theta} = \mathbb{E}_\theta [Q_T \text{score}_{X,\theta}].$$

All expectations here are finite by the Cauchy–Schwarz inequality, because $\mathbb{E}_\theta Q_T^2 < \infty$ (conditional Poisson moments are polynomials in the λ_j^X) and $\text{score}_{X,\theta} \in L^2$. Conditioning on N_T and using (A.3) gives $\mathbb{E}_\theta [Q_T \text{score}_{X,\theta}] = \mathbb{E}_\theta [Q_T \text{score}_{N,\theta}]$, and (A.4) follows from $\mathbb{E}_\theta \text{score}_{N,\theta} = 0$. $\square$

Lemma A.2 (Augmented experiment). *Let $(R_\theta)_{\theta \in \Theta}$ be real, even functions such that $\theta \mapsto R_\theta$ is C^1 from Θ into $A(\mathbb{T})$ and $r_- \leq R_\theta(\omega) \leq r_+$ for constants $0 < r_- \leq r_+ < \infty$; for example (5.1), with $r_- = 1 + m$ and $r_+ = 1 + M$. Let $Z = Z^{(T)} \sim N(0, \text{Toe}_T(R_\theta))$ and, conditional on $Z = z$, let the $N_{T,j}$, $j \in J_T$, be independent with*

$$N_{T,j} \mid Z = z \sim \text{Pois}(\lambda_j(\theta, z)), \quad \lambda_j(\theta, z) = |(F_T L_T(\theta) z)_j|^2 = \lambda_j^X(L_T(\theta) z),$$

where $L_T(\theta)$ is given by (5.2). Then for every $\theta \in \Theta$ the law of N_T is the original one, and

$$I_T(\theta_0) \leq I_T^Z + \mathbb{E}_{\theta_0} I_T^{N|Z}(Z).$$

Here

$$I_T^Z = \frac{1}{2} \text{tr}\{\text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R}) \text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R})\}, \quad I_T^{N|Z}(z) = \sum_{j \in J_T} \frac{\dot{\lambda}_j(z)^2}{\lambda_j(z)}$$

are the Fisher information of Z and the conditional Fisher information of N_T given $Z = z$, both at θ_0 , with $\lambda_j(z) = \lambda_j(\theta_0, z)$ and $\dot{\lambda}_j(z) = \partial_\theta \lambda_j(\theta, z)|_{\theta_0}$.

Proof. Law of N_T . Both square roots in (5.2) are real symmetric, so $L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta)$. Hence $L_T(\theta) Z \sim N(0, \text{Toe}_T(S_\theta))$, and the $\lambda_j(\theta, Z)$ are the intensities in (2.1) computed from $X^{(T)} = L_T(\theta) Z$.

Smoothness. By (3.1), $r_- I \leq \text{Toe}_T(R_\theta) \leq r_+ I$. Since $x^{1/2}$ and $x^{-1/2}$ are analytic on neighbourhoods of the relevant positive spectral intervals, standard analytic matrix functional calculus [6, Chapter 3] implies that the maps $\theta \mapsto \text{Toe}_T(S_\theta)^{1/2}$ and $\theta \mapsto \text{Toe}_T(R_\theta)^{-1/2}$ are C^1 in operator norm, hence so is the invertible matrix $L_T(\theta)$. Consequently $\lambda_j(\theta, z)$ is a real quadratic form in z with C^1 coefficients, and on a compact neighbourhood $\Theta_1 \subset \Theta$ of θ_0 ,

$$\lambda_j(\theta, z) + |\partial_\theta \lambda_j(\theta, z)| \leq C \|z\|^2, \quad \theta \in \Theta_1.$$

Differentiation of the marginal. The joint law of (Z, N_T) has density $\pi_\theta(z, n) = \psi_\theta(z) K_\theta(n \mid z)$ with respect to Lebesgue measure times counting measure, where ψ_θ is the $N(0, \text{Toe}_T(R_\theta))$ density and

$$K_\theta(n \mid z) = \prod_{j \in J_T} k_{n_j}(\lambda_j(\theta, z)), \quad k_\nu(\lambda) = e^{-\lambda} \frac{\lambda^\nu}{\nu!}.$$

For $\nu \geq 1$, $k'_\nu(\lambda) = k_{\nu-1}(\lambda) - k_\nu(\lambda)$, while $k'_0(\lambda) = -k_0(\lambda)$. Since $0 \leq k_\nu \leq 1$, it follows that $|k'_\nu| \leq 2$ on $[0, \infty)$. Therefore $|\partial_\theta K_\theta(n \mid z)| \leq 2 \sum_{j \in J_T} |\partial_\theta \lambda_j(\theta, z)| \leq C \|z\|^2$ for every fixed n . Together with the analogue of (A.2) for ψ_θ (with M replaced by r_+), this bounds $|\partial_\theta \pi_\theta(z, n)|$, uniformly on Θ_1 , by $C(1 + \|z\|^2) e^{-\|z\|^2/(2r_+)}$. Hence, for each fixed n , $p_\theta(n) = \int \pi_\theta(z, n) dz$ may be differentiated

under the integral sign. No interchange of differentiation with the infinite sum over n is used or needed here.

Scores. For Lebesgue-a.e. z , all $\lambda_j(\theta_0, z)$, $j \in J_T$, are positive, because $L_T(\theta_0)$ is invertible. At such z ,

$$\partial_\theta \log \pi_\theta(z, n)|_{\theta_0} = \text{score}_Z(z) + \text{score}_{N|Z}(z, n), \quad \text{score}_{N|Z}(z, n) = \sum_{j \in J_T} (n_j - \lambda_j(z)) \frac{\dot{\lambda}_j(z)}{\lambda_j(z)},$$

where $\text{score}_Z = \partial_\theta \log \psi_\theta|_{\theta_0}$. The previous step therefore gives

$$\dot{p}_{\theta_0}(n) = \int \pi_{\theta_0}(z, n) \{ \text{score}_Z(z) + \text{score}_{N|Z}(z, n) \} dz,$$

that is,

$$\text{score}_{N, \theta_0}(N_T) = \mathbb{E}_{\theta_0} [\text{score}_Z(Z) + \text{score}_{N|Z}(Z, N_T) \mid N_T].$$

Information. Conditional on $Z = z$, $\text{score}_{N|Z}$ has mean zero and second moment $I_T^{N|Z}(z)$. With $\check{X} = F_T L_T(\theta_0) Z$ and $V = F_T \dot{L}_T Z$ we have $\dot{\lambda}_j = 2 \text{Re}(\check{X}_j^* V_j)$, so $\dot{\lambda}_j^2 / \lambda_j \leq 4 |V_j|^2$ and $\mathbb{E} I_T^{N|Z}(Z) < \infty$. Hence $\text{score}_Z + \text{score}_{N|Z} \in L^2$, the cross term vanishes, and $\mathbb{E}(\text{score}_Z + \text{score}_{N|Z})^2 = I_T^Z + \mathbb{E} I_T^{N|Z}(Z)$. Conditional Jensen applied to the previous display completes the proof. $\square$

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